

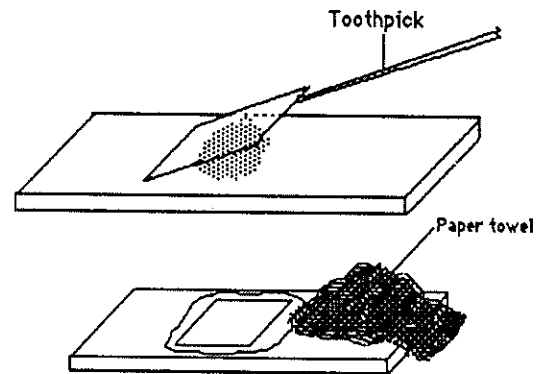
MICROSCOPE ACTIVITY 2 PREPARING A WET MOUNT

Aim: To prepare a wet mount.

Things I Need: Microscope, microscope lamp, tooth pick, pair of scissors, dropper bottle of water, paper towel, 2 microscope slides and cover slips, piece of news paper, colour magazine picture.

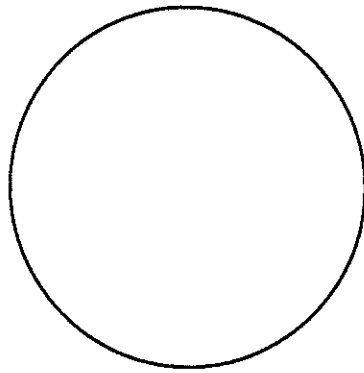
Procedure:

1. Cut out the letter "e" from the news paper, and a small square from the coloured picture.
2. Place each on a separate slide and add a drop of water onto each piece of paper.
3. For each slide, place the cover slip as shown in the diagram opposite, and gently lower until it is flat.
4. Use a piece of paper towel to soak up excess water.
5. Set up the microscope, according to the rules, and view the slides.

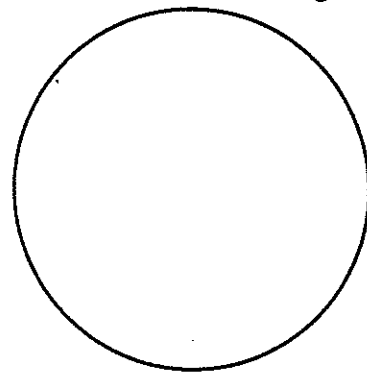


Answer the following questions relating to the above activity.

1. In the the space below, draw the letter "e" as seen under low power.



2. In the the space below, draw the letter "e" as seen under high power.



2. Compare the appearance of the letter "e" when viewed under the microscope and when seen with the 'naked' eye.

3. Describe the appearance of the coloured picture under low power.

4. Describe the appearance of the coloured picture under high power.

5. Compare the appearance of the coloured picture when viewed under the microscope and when seen with the 'naked' eye.

6. When looking into the eyepiece and:
 - (a) moving the slide away from me, the object in the field of view moves _____
 - (b) moving the slide towards me, the object in the field of view moves _____
 - (c) moving the slide to the right, the object in the field of view moves _____
 - (d) moving the slide to the left, the object in the field of view moves _____
7. If a fast moving microscopic water animal in the field of view moves to the right, then in which direction must the slide be moved so that the animal remains in view? _____
8. Describe the appearance of air bubbles trapped under the cover slip. _____

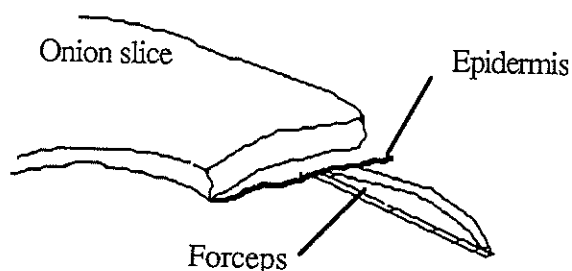
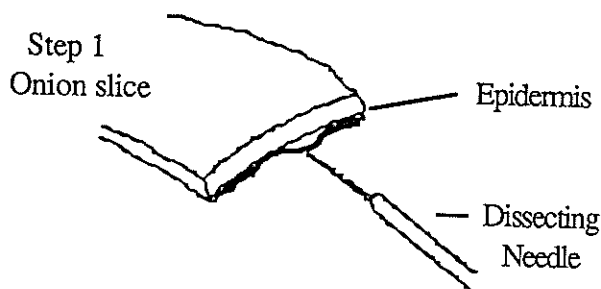
MICROSCOPE ACTIVITY 3 ONION EPIDERMIS

Aim: To observe an Onion Epidermis.

Things I Need: Microscope, microscope lamp, tooth pick, dropper bottle of water, paper towel, microscope slides and cover slips, gem blade, forceps, dissecting needle, slice of onion.

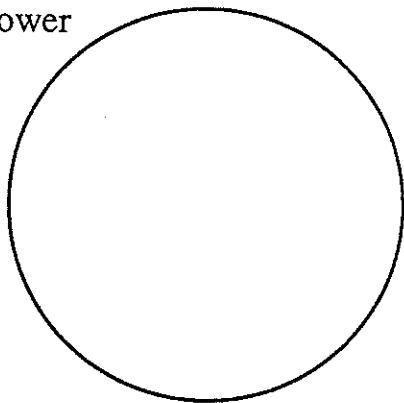
Procedure:

1. Cut a slice of onion, being careful to cut through only one slice of the layers.
2. The epidermis is the very thin membrane on the underside of the onion layer. Using the dissecting needle tease the epidermis away from the onion layer and use the forceps to pull the epidermis away as shown in the diagrams opposite.
3. Very carefully place a piece of the onion epidermis on a clean slide, making sure that it is flat. Add a drop of water and cover with a cover slip.
4. Set up the microscope.

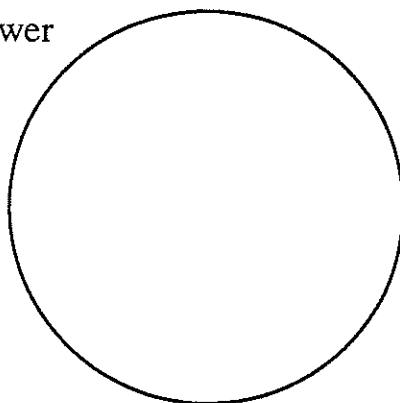


5. In the space below draw a neat diagram of the onion epidermis under:

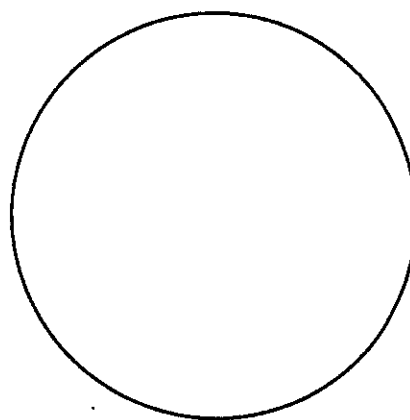
Low Power



High Power



6. Examine an onion epidermis cell under high power and draw a diagram of what is observed in the space opposite.
7. What are some of the structures within the cell?



MICROSCOPE ACTIVITY 4 CHEEK CELLS

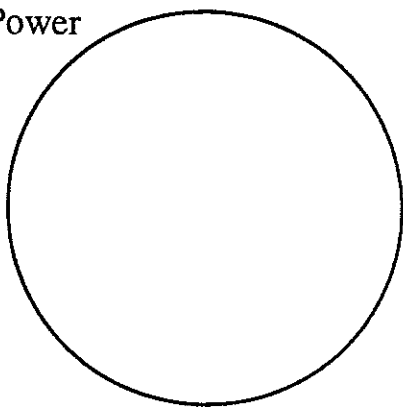
Aim: To observe cheek cells.

Things I Need: Microscope, microscope lamp, tooth pick, dropper bottle of methylene blue, paper towel, microscope slide and cover slip.

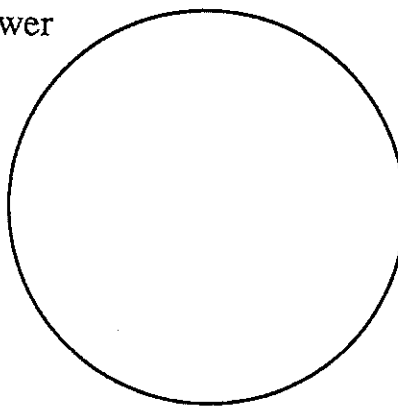
Procedure:

1. Using the thick end of a tooth pick, scrape the inside of my cheek.
2. Smear the the end of the tooth pick over the centre of the slide.
3. Add 1 drop of methylene blue, and cover with the cover slip. Remove any excess stain with the edge of some paper towel.
4. Set up the microscope.
5. Examine the slide for cheek cells under low power. If there are no cells, prepare a new slide, this time scraping the cheek a little harder.
6. In the space below draw a neat diagram of the cheek cells under:

Low Power



High Power



7. Describe the cell structure.

8. On the diagrams on the previous page, label the parts of the cells.

9. What are the differences between a human cheek cell and the onion epidermis cell?

MICROSCOPE ACTIVITY 5 MINI GRID

Aim: To observe the size of things.

Things I Need: Microscope, microscope lamp, microscope slide and cover slip, mini grid or 1 square centimetre of graph paper (1 mm lines).

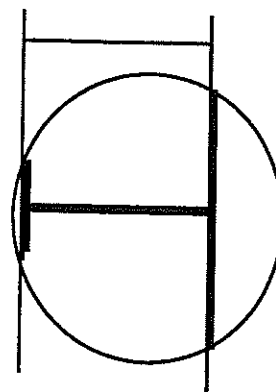
Procedure:

- Calculate the magnification of low and high powers obtained by using the formula below. Record the results in the following table.

MAGNIFICATION = EYE PIECE MAGNIFICATION x OBJECTIVE MAGNIFICATION

	Eyepiece Magnification	Objective Magnification	Microscope Magnification	Diameter of Field of View
Low Power				
High Power				

- Set up the microscope on low power with the mini grid on the stage.
- Move the mini grid on the stage so that one of the block lines is on the edge of the field of view, as shown in the diagram opposite.
- Estimate the diameter of the field of view by counting the number of block lines seen through the microscope, and record the results in the table above.



- Calculate the diameter of the field of view under high power by using the following formula:

$$\text{Diameter Field of View} = \frac{\text{Magnification of Low Power} \times \text{Diameter of Field of View of Low Power}}{\text{Magnification of High Power}}$$